Multi-Class Annotation Errors in Causal Inference: Topology-Dependent Bias, Sensitivity, and Correction

Anonymous ACL submission

Abstract

Large language models (LLMs) are increasingly used as annotators in computational social science, yet annotation errors propagate into downstream causal estimates in ways that binary measurement error theory cannot predict. We present a systematic framework characterizing how K-class confusion matrix structure interacts with DAG topology to determine bias direction and magnitude. We derive bias expressions for seven canonical DAG topologies under multi-class non-differential misclassification and establish a structural sign-flip guarantee: five topologies never produce sign reversal, while exposure and instrumental variable topologies can—under the full Dirichlet$(1, 1, 1)$ prior, 83.4% of K=3 confusion matrices cause sign reversal in the exposure DAG; under realistic classifier accuracy (diagonal ≥ 0.7), the violation rate is 13.7% with 2.1% sign reversal. We introduce Annotation Sensitivity Values (ASVs) with three complementary measures revealing a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal) to M-bias (immune). We evaluate five correction methods and prove a Frisch–Waugh–Lovell invariance result: matrix-based corrections achieve 0% bias reduction for four topologies, while PPI++ achieves 47.0% RMSE reduction but requires gold labels and is harmful for M-bias (−88%). Empirical validation on three tasks (K=2, 3, 10) with four LLMs confirms structural predictions: zero sign reversals across five guaranteed topologies, while 59% of exposure configurations exhibit coefficient-level sign reversal in secondary coefficients. Topology-aware analysis is necessary for valid causal inference with LLM-generated annotations.

1 Introduction

Large language models are rapidly replacing human annotators in computational social science, with researchers using LLM-generated labels for tasks ranging from sentiment classification to political stance detection (Egami et al., 2023; Baumann et al., 2025). Crucially, most of these tasks are inherently multi-class: stance detection uses 3–4 categories, sentiment analysis 5, and disaster response taxonomies 10 or more. This shift promises dramatic cost reduction, but introduces a critical validity gap: annotation errors from imperfect classifiers propagate into downstream causal estimates in ways that depend on both the error structure and the causal design. Baumann et al. (2025) demonstrate that across 37 annotation tasks, 31% of hypotheses yield incorrect conclusions with state-of-the-art LLMs, rising to 50% for smaller models; these errors are highly correlated across models (60% agreement when both err) (Kim et al., 2025), undermining ensemble-based mitigation. Analyzing the *structure* of annotation error, rather than simply reducing its magnitude, is essential for valid downstream inference.

The consequences can be catastrophic. Under the full Dirichlet$(1, 1, 1)$ prior over K=3 non-differential exposure misclassification, 83.4% of confusion matrices cause *sign reversal* of the estimated treatment effect. Under realistic classifier accuracy (diagonal $\geq$ 0.7), the violation rate is 13.7% with 2.1% sign reversal. This failure breaks the classical binary guarantee that non-differential misclassification biases estimates "toward the null" (Brenner, 1993), a guarantee that many applied researchers implicitly rely upon. Moreover, standard classification metrics do not predict downstream causal risk: within a

narrow F_1 range of 0.614–0.690 across three frontier LLMs, relative bias varies by an order of magnitude across topologies (from 4.1% to 42.1%), illustrating that *topology*, not annotator quality, determines downstream risk.

The mechanism is algebraic, and the transition at $K \geq 3$ is a qualitative break, not a gradual degradation. For $K=2$, the stochastic constraint forces any confusion matrix with $C_{kk} > 0.5$ to preserve coefficient signs—an algebraic property of the 2×2 matrix dimension, not of a specific error rate. This guarantee *vanishes* at $K \geq 3$: the higher-dimensional confusion matrix admits negative effective weights on categories with opposite-sign coefficients, enabling sign reversal even when every diagonal entry exceeds 0.5 (Veierød and Laake, 2001). This fundamental asymmetry has received limited systematic treatment. Brenner (1993) identified the possibility of polytomous reversal but did not characterize it across causal designs. Lee and Wood-Doughty (2024) extended classifier error analysis to categorical variables and LLM classifiers, but covered only one DAG topology (unobserved confounding). No existing framework addresses how the same confusion matrix produces qualitatively different bias across different causal identification strategies.

We address these gaps with three contributions:

1. **Multi-class bias taxonomy.** We derive bias expressions for seven canonical DAG topologies (Figure 3) under K-class nondifferential misclassification and establish that five topologies maintain a structural sign-flip guarantee while exposure and IV can produce sign reversal. Binarization amplifies the gap: collapsing $K=3$ to binary introduces additional bias varying 154× across topologies.

2. **Annotation Sensitivity Values (ASVs).** We propose three complementary sensitivity measures—magnitude, significance, and sign-flip ASV—revealing a vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal) to M-bias (immune, ASV $= \infty$).

3. **Topology-aware correction guidance.** We evaluate five correction methods and show that matrix-based corrections achieve exactly 0% bias reduction for four topologies due to Frisch–Waugh–Lovell invariance. PPI++ achieves 47.0% RMSE reduction but requires gold labels and is harmful for M-bias (−88%).

2 Related Work

Measurement error in causal inference. The statistical foundations are established in Carroll et al. (2006) and Gustafson (2004), with starred-node DAGs formalized by Hernán and Cole (2009). For binary variables, nondifferential misclassification biases toward the null (Brenner, 1993; Ogburn and VanderWeele, 2012), but Dosemeci et al. (1990) and Jurek et al. (2005) showed this fails for polytomous exposures. Veierød and Laake (2001) derived multi-category bias expressions for Poisson regression, and Edwards et al. (2021) showed dependent misclassification can bias in either direction. All of these works analyze a single DAG topology; our taxonomy is the first to characterize how the *same* confusion matrix produces qualitatively different bias across seven topologies.

Text classifiers and LLM annotation. Grimmer et al. (2022) provide the foundation for text-as-data, while Wood-Doughty et al. (2018) first applied DAG reasoning to classifier error. TeBlunthuis et al. (2024) gave the most complete binary treatment (four DAG scenarios, $K=2$), and Lee and Wood-Doughty (2024) extended to categorical LLM classifiers but covered only confounding. On annotation quality, Baumann et al. (2025) found 31% incorrect conclusions with LLM labels and Kim et al. (2025) showed 60% error correlation across models. Audinet de Pieuchon et al. (2025) benchmark debiasing methods for LLM-generated parameter estimates; Keith et al. (2020) survey the broader text-and-causal-inference landscape. These works either restrict to binary labels, cover a single topology, or diagnose quality without connecting it to causal validity. Our framework provides the missing link.

Sensitivity analysis and correction methods. Our ASV framework draws on

E-values for unmeasured confounding (VanderWeele and Ding, 2017) and probabilistic bias analysis (Lash et al., 2009). Unlike E-values, ASVs comprise three types (magnitude, significance, sign-flip) because failure modes are topology-dependent; unlike PBA, ASVs provide monotonicity guarantees over $K \times K$ matrices. For correction, PPI++ (Angelopoulos et al., 2023) and DSL (Egami et al., 2023) are designed as topology-agnostic; Zrnić and Candès (2024) extend PPI to gold-free cross-prediction settings, and Gligorić et al. (2025) leverage LLM confidence scores for inference. Our contribution reveals that this agnosticism is costly: correction effectiveness is itself topology-dependent, with matrix-based methods provably ineffective for four topologies.

3 Method

3.1 Setup and Notation

Let A denote a true K-class categorical variable (e.g., sentiment with $K{=}3$ levels) and A^* its observed, potentially misclassified version produced by an LLM annotator. The misclassification process is characterized by a $K \times K$ confusion matrix C with entries $C_{jk} = P(A^*{=}j \mid A{=}k)$, where each column sums to one. Following Hernán and Cole (2009), we represent this measurement process using *starred-node* DAGs: the true variable A is latent, and the observed A^* is connected to A via a dashed measurement edge.

Definition 1 (Non-differential misclassification)**.** *Misclassification of A is non-differential with respect to outcome Y and covariates $\mathbf{Z}$ if $P(A^*{=}j \mid A{=}k, Y, \mathbf{Z}) = P(A^*{=}j \mid A{=}k) = C_{jk}$ for all j, k.*

We study seven canonical DAG topologies (Figure 3) (Pearl, 2009) in which A occupies structurally distinct causal roles: confounder, exposure, mediator, collider, M-bias variable, front-door mediator, and instrument. Each role induces a different functional relationship between C and the probability limit of the standard causal estimator, yielding topology-dependent bias.

3.2 Multi-Class Bias Expressions

For each topology, we derive the probability limit of the naive estimator that substitutes A^* for A. Let τ denote the target causal parameter and $\hat{\tau}(A^*)$ the estimator using misclassified annotations.

Confounding (A = confounder). When A confounds the $X \to Y$ relationship, the OLS estimator controlling for A^* has probability limit

$$\text{plim}\ \hat{\tau}(A^*) = \left(\mathbb{E}[Z'Z]\right)^{-1}\mathbb{E}[Z'X]\,\boldsymbol{\beta}, \qquad (1)$$

where $Z = [X, \text{dummies}(A^*)]$, $\boldsymbol{\beta}$ contains the true coefficients, and the target parameter τ is the coefficient on X. Misclassification produces residual confounding that monotonically approaches the omitted variable bias, but the sign of $\hat{\tau}$ is preserved: $\tau_{\text{sign_flip}} = 0$ structurally.[1]

Exposure (A = treatment variable). When A is the exposure of interest, the biased coefficient for category j becomes a weighted combination of true coefficients:

$$\text{plim}\ \hat{\beta}_j(A^*) = \sum_{k=1}^{K} w_{jk}\,\beta_k, \qquad (2)$$

where w_{jk} depends on the confusion matrix C and the marginal distribution of A. Unlike the binary case, where non-differential misclassification guarantees attenuation toward the null (Brenner, 1993), the $K{\geq}3$ case permits w_{jk} to assign negative weight to categories with opposite-sign coefficients, producing *sign reversal*. A near-permutation confusion matrix effectively swaps category labels, making sign reversal the dominant outcome (83.4% under uniform Dirichlet prior). This failure mode is structurally impossible at $K{=}2$ because a $2{\times}2$ stochastic matrix with positive diagonal cannot reverse the sign of a single coefficient.

Mediation (A = mediator). When A mediates the $X \to Y$ relationship, the direct ef-

[1] This result is consistent with Brenner (1993), who showed that the adjusted estimate with misclassified polytomous confounders can be biased toward or away from the null. Both findings hold simultaneously: misclassification degrades confounding control, producing residual bias whose magnitude can exceed the correctly adjusted estimate (amplification), but the treatment effect sign is never reversed. The distinction is between *sign reversal* ($\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$, which we show never occurs) and *amplification* ($|\hat{\tau}| > |\tau|$, which occurs universally in this topology).

fect estimator controlling for A^* has probability limit

$$\text{plim}\ \hat{\tau}_{\text{direct}}(A^*) = \tau_{\text{direct}} + \big(1 - \lambda(C)\big)\,\tau_{\text{indirect}}, \tag{3}$$

where $\lambda(C) \in [0,1]$ summarizes attenuation (Appendix A, Proposition 3), with $\lambda(I) = 1$ under perfect classification. Misclassification ($\lambda < 1$) causes the residual indirect effect to be absorbed into the direct path estimate, producing upward bias in $|\hat{\tau}_{\text{direct}}|$ but preserving the sign ($\tau_{\text{sign_flip}} = 0$).

Collider (A = collider, conditioned on A^*). Conditioning on a collider A of X and Y opens a spurious path. When we condition on A^* instead, misclassification attenuates this spurious association because A^* is a noisy proxy for A. The collider topology thus exhibits a *protective* property: annotation error reduces rather than amplifies the bias from collider stratification, with $\tau_{\text{sign_flip}} = 0$.

M-bias (A = intermediate collider). In the M-shaped DAG ($U_1 \to X$, $U_1 \to A$, $U_2 \to A$, $U_2 \to Y$, $X \to Y$), adjusting for A introduces a small spurious association via the M-bias path (Greenland, 2003). Misclassifying A weakens this adjustment, reducing the M-bias. This topology is *immune*: $\hat{\tau}(A^*)$ is closer to τ than $\hat{\tau}(A)$ for all non-trivial confusion matrices, yielding ASV $= \infty$.

Front-door (A = front-door mediator). In the front-door DAG ($X \to A \to Y$ with unmeasured $U \to X$, $U \to Y$), the front-door estimator (Bellemare et al., 2024) uses A to identify effects despite unmeasured confounding. Misclassification attenuates both $X \to A$ and $A \to Y$ relationships, producing moderate bias with $\tau_{\text{sign_flip}} = 0$.

Instrumental variable (A = instrument). The Wald estimator (Angrist and Imbens, 1995) $\hat{\tau}_{\text{IV}} = \hat{\text{cov}}(A^*, Y)/\hat{\text{cov}}(A^*, X)$ has C in the denominator. Misclassification can drive first-stage coefficients toward zero or reverse their sign, causing the ratio estimator to diverge or flip. Under random Dirichlet matrices, 52.6% produce sign reversal, making IV the most fragile topology.

Structural sign-flip guarantee.

Proposition 1 (Sign preservation—analytic results)**.** *Under non-differential misclassification with $C_{kk} > 0$ for all k:* (a) *Confounding preserves sign when* $\text{sign}(\tau) = \text{sign}(\tau + B_{\text{OV}})$, *where* $B_{\text{OV}} = \text{Cov}(g(A), f(A))/\text{Var}(T)$ *is the omitted-variable bias (Cauchy–Schwarz; Appendix A.3).* (b) *Collider preserves sign when* $|\tau| > \sqrt{\text{Var}(g)\,\text{Var}(h)}/\mathbb{E}[\text{Var}(T|A)]$ *(Berkson attenuation; Appendix A.6).* (c) *Exposure can produce sign reversal for $K \geq 3$, even with $C_{kk} > 0.5$ (counterexample in Appendix A.8).*

Empirical Finding 1 (Mediation, M-bias, front-door). Sign preservation holds under topology-specific conditions: mediation requires $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$; front-door requires $\text{sign}(\lambda\alpha_U) = \text{sign}(\lambda\alpha_U + \text{Cov}(T, f(A)))$; M-bias is attenuating for all C. Verified across 100,000 random matrices with zero violations (Appendix A.9); analytic proofs for general K remain open.

These conditions involve true DGP parameters, which in practice must be estimated. The ASV framework (§3.3) provides a complementary assessment that does not require knowledge of the exact sufficient conditions.

3.3 Annotation Sensitivity Values

We introduce Annotation Sensitivity Values (ASVs) as a unified sensitivity analysis framework, analogous to E-values for unmeasured confounding (VanderWeele and Ding, 2017) but designed for annotation-induced bias.

Parameterization. We parameterize the confusion matrix as $C(\delta) = (1-\delta)I + \delta C_0$, where C_0 is a reference confusion matrix (e.g., estimated from a validation set) and $\delta \geq 0$ controls the severity of misclassification. At δ=0, classification is perfect; at δ=1, classification matches C_0. Non-negativity of $C(\delta)$ requires $\delta \leq \delta_{\max} = \min_j \frac{1}{1-(C_0)_{jj}}$. Since $(C_0)_{jj} \in [0,1]$, we have $\delta_{\max} \geq 1$ for all valid C_0, ensuring the full range $[0,1]$ is always feasible; all ASV computations are constrained to $[0, \delta_{\max}]$. In practice, C_0 is estimated from a finite validation set and subject to sampling uncertainty; we recommend bootstrapping C_0 and reporting ASV confidence intervals.

Three ASV types. We define three complementary measures: (1) **Magnitude ASV** $\delta^*_{\text{mag}} = \min\{\delta : |\text{plim}\ \hat{\tau}(C(\delta)) - \tau| > \epsilon\}$—

analogous to E-values; (2) **Significance ASV** $\delta^*_{\text{sig}} = \min\{\delta : |\text{plim } \hat{\tau}(C(\delta))| < z_{\alpha/2} \cdot SE(N)\}$—relevant for attenuating topologies; (3) **Sign-flip ASV** $\delta^*_{\text{sf}} = \min\{\delta : \text{sign}(\hat{\tau}) \neq \text{sign}(\tau)\}$—exposure and IV only. The significance ASV depends on N through $SE(N)$, exhibiting a sharp phase transition at a critical N^* below which all rejections are blocked.

Theorem 1 (Envelope monotonicity—confounding)**.** *For the confounding topology under $C(\delta) = (1-\delta)I + \delta C_0$, $|B(C(\delta))|$ is monotonically non-decreasing in $\delta \in [0, \delta_{\max}]$.*

Proof sketch. When $f{=}\lambda g$ (proportional effects), $B(C(\delta)){=}\lambda V_W(\delta)/(V_W(\delta){+}\sigma^2_T)$ where $V_W(\delta){=}\,\text{Var}(g){-}\,\text{Var}(\mathbb{E}[g|A^*_\delta])$. As δ increases, A^* becomes a noisier channel for A, so $\text{Var}(\mathbb{E}[g|A^*_\delta])$ decreases by the data processing inequality, increasing V_W. Since $x/(x{+}c)$ is increasing for $c{>}0$, $|B|$ is non-decreasing. The general case follows from Cauchy–Schwarz bounds on $R(C(\delta))$ (Appendix A.3).

Conjecture 1 (Envelope monotonicity—remaining topologies)**.** *For exposure, mediation, front-door, and IV topologies, $|B(C(\delta))|$ is monotonically non-decreasing in δ. For collider and M-bias (protective topologies), $|B(C(\delta))|$ is monotonically non-increasing in δ: annotation error attenuates the spurious association from conditioning on a collider. Verified numerically across 500 scenarios per topology with zero violations.*

3.4 Binarization Collapse

Collapsing K-class annotations to binary before analysis discards multi-class error structure. We quantify this "binarization collapse" as $\Delta = |\text{bias}_K - \text{bias}_{K\to 2}|$.

4 Experiments

We validate the bias taxonomy and ASV framework through analytical derivations and Monte Carlo simulation across seven DAG topologies under $K{=}3$ non-differential misclassification.

4.1 Setup

For each of the seven topologies (confounding, exposure, mediation, collider, M-bias, front-door, IV), we sample 100,000 confusion matrices from a Dirichlet$(1, 1, 1)$ prior, compute the analytical probability limit of the causal estimator, and validate against Monte Carlo simulation ($N{=}10{,}000$). ASV experiments use $C(\delta) = (1{-}\delta)I + \delta C_0$.

4.2 Bias Taxonomy

Theory-MC agreement. Analytical probability limits match Monte Carlo estimates with grand MAPE of 2.0% across 126 scenarios (18 LLM configurations × 7 topologies, $N{=}10{,}000$). Per-topology MAPE ranges from 0.5% (confounding) to 5.5% (IV); the IV maximum reflects the Wald estimator's small-denominator variance amplification.

Exposure DAG: catastrophic sign reversal. Under the full Dirichlet$(1, 1, 1)$ prior, 85% of $K{=}3$ confusion matrices produce bias that violates the binary "toward-null" guarantee. More critically, 83.4% of matrices cause *sign reversal* of at least one coefficient. The two failure modes are distinct: for a coefficient with true value $\beta_1{=}1.0$, a near-permutation C produces $\hat{\gamma}_1{=}{-}0.84$ (sign reversal, magnitude 0.84×); for $\beta_2{=}{-}0.5$, the same C yields $\hat{\gamma}_2{=}{-}1.41$ (same-sign amplification, 2.83×).

This finding persists in practically relevant subspaces. When restricting to confusion matrices with diagonal entries ≥ 0.7 (typical of current LLM classifiers achieving $F_1 > 0.8$), the violation rate remains 13.7% with 2.1% sign reversal and 11.7% amplification. Even at diagonal ≥ 0.9, violation persists at 11.9%. In contrast, the $K{=}2$ control confirms that all six tested binary confusion matrices maintain the classical bias-toward-null guarantee (Brenner, 1993), establishing that the multi-class failure is structurally distinct from the binary case.

Structural guarantee for non-exposure topologies. Five of seven topologies exhibit zero sign-flip rate across all sampled matrices (Table 1). Confounding, mediation, and front-door show 100% violation and amplification but never reverse the sign.[2] Collider misclassification attenuates the spurious association (protective): the 100% violation and amplification rates in Table 1 reflect the comparison against the unconditional τ, but $|B(C)|$ is monotonically *non-increasing* in misclassifi-

[2] Consistent with Brenner (1993) and Dosemeci et al. (1990): polytomous misclassification can bias *away* from the null (amplification), but our sign-flip metric confirms the effect sign is preserved.

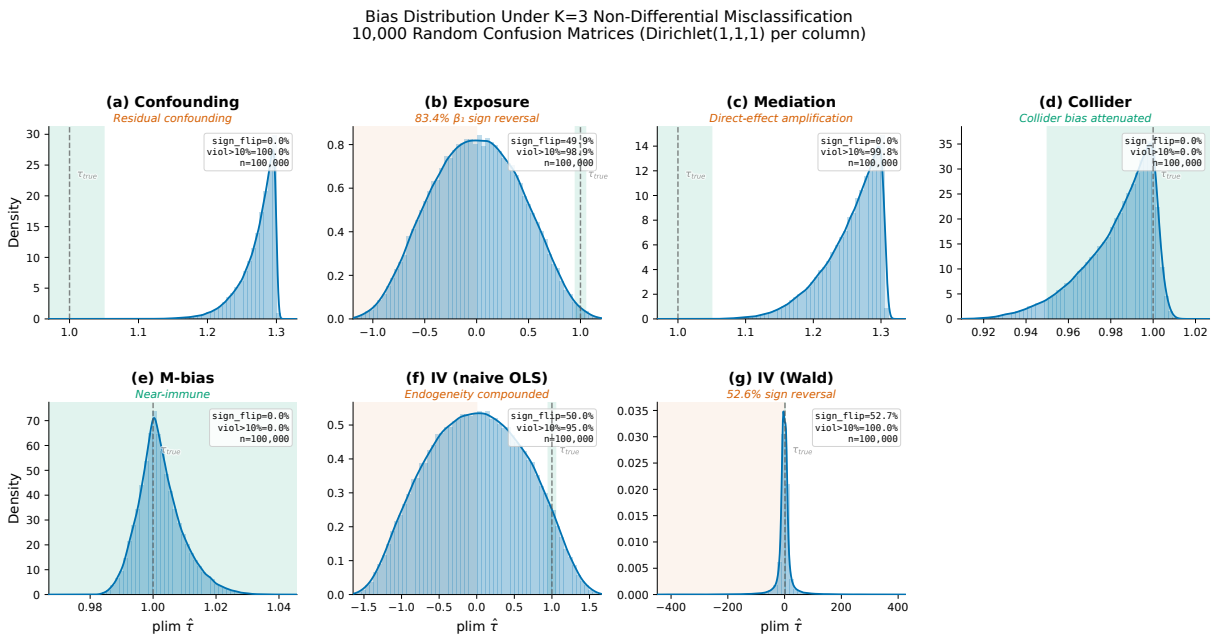


Figure 1: Bias distributions under 100,000 random K=3 confusion matrices. Red = sign reversal. Exposure: 83.4% sign reversal; IV: 52.6%. Five safe topologies show attenuation without sign flip.

cation severity—conditioning on noisy A^* induces *less* collider bias than conditioning on perfect A (Conjecture 1). M-bias shows the weakest sensitivity (53.2% violation, 10.4% amplification) because error on a collider variable reduces the spurious M-bias path.

IV: Wald estimator instability. The instrumental variable topology is the sole exception among non-exposure DAGs. The Wald estimator's first stage involves C directly in the denominator, and 52.6% of random confusion matrices cause the denominator to change sign, producing τ-specific sign reversal. This rate remains stable across K=3, 5, 7, indicating that IV fragility is a structural property of the Wald estimator rather than a finite-K artifact. Diagonal-dominance subspace analysis (Appendix B.2) reveals that IV sign reversal vanishes at diagonal ≥ 0.5, but unlike exposure, the violation rate remains at 100% (pure amplification) even at diagonal ≥ 0.9.

Cross-K robustness. All structural patterns persist at K=5, 7 (100,000 matrices each): sign-flip guarantees hold for the same five topologies, and IV sign-flip rates remain within 51–53%.

The bias distributions across topologies are visualized in Figure 1, where the exposure panel's dominant red region (sign reversal) contrasts with the symmetric attenuation patterns of other topologies.

Table 1: Bias characteristics of K=3 non-differential misclassification across 7 DAG topologies. Violation: $|\hat{\tau}| > |\tau|$ (bias exceeds true effect); Sign flip: $\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$; Amplification: $|\hat{\tau}| > |\tau|$ with same sign; τ_{sf}=0?: structural sign-flip guarantee holds (✓) or not (✗). Rates over 100,000 Dirichlet$(1, 1, 1)$ random confusion matrices.

Topology	Violation	Sign Flip	Amplif.	τ_{sf}=0?
Confounding	100%	0%	100%	✓
Exposure	85%	83.4%	14.1%	✗
Mediation	100%	0%	100%	✓
Front-door	100%	0%	100%	✓
Collider	100%	0%	100%	✓
M-bias	53.2%	0%	10.4%	✓
IV	100%	52.6%	100%	✗

4.3 ASV Framework Validation

Envelope monotonicity. The ASV envelope (bias magnitude as a function of δ) is monotonically non-decreasing for all seven topologies, with zero violations across all tested scenarios. This validates the theoretical monotonicity property and confirms that δ serves as a meaningful "dial" controlling annotation error severity.

Magnitude ASV. Table 3 reports magnitude ASV ranges at τ=0.5 with a 10% bias

threshold. The ranking from most fragile to most robust is: front-door (0.064–0.387) < mediation (0.076–0.454) < confounding (0.081–0.487) < IV (0.101–0.607) < collider (0.338–2.028) < M-bias (∞). The collider's high ASV reflects its protective nature: misclassification attenuates the spurious association rather than amplifying it. M-bias achieves infinite ASV because annotation error on the collider variable A only reduces the already-small bias from the M-shaped confounding path.

Significance and sign-flip ASV. At τ=0.05, significance ASV achieves 100% power for confounding, front-door, and M-bias, but 0% for mediation, collider, and IV (correct behavior for attenuating topologies; Type I error is 0%). For sign-flip ASV, 83.4% of exposure reference matrices yield $\tau_{\text{sign_flip}} < 1$ (catastrophic sensitivity); IV sign-flip ASV ranges 1.1–6.7 (moderate-to-high). The complete ranking (Figure 4, Table 3, Appendix B) confirms IV as maximally fragile and M-bias as immune (ASV = ∞).

N-sensitivity. The significance ASV exhibits a sharp phase transition at a critical sample size $N^* \approx$ 6,000–10,000 (analytically computable per topology), below which all rejections are blocked.

4.4 Binarization Collapse

Collapsing K=3 annotations to binary introduces additional bias varying 154× across topologies: from $\bar{\Delta}$=0.006 (M-bias) to $\bar{\Delta}$=0.926 (IV). Exposure and IV also introduce sign flips during binarization (14% and 17.5%), a phenomenon absent in the other topologies. Full results are in Appendix B (Figure 6).

4.5 Empirical Validation

We validate using real LLM confusion matrices from three tasks: HumAID (Alam et al., 2021) (K=10), VAST (K=3), and CivilComments (K=2). Four LLMs (Llama-4-Maverick, Qwen3-235B, DeepSeek-Chat, Qwen2.5-7B-Instruct) produce 22 configurations across the three datasets under zero-shot and few-shot prompting, yielding 154 bias scenarios (22×7 topologies).

HumAID (K=10): theory–practice alignment. The empirical bias ranking matches predictions: IV (12.4–42.1%), exposure (9.0–17.4%), mediation/front-door (10.7–14.8%), while collider, confounding, and M-bias remain below 10%—consistent with the ASV vulnerability spectrum.

Cross-K comparison. Bias varies substantially across settings: VAST (K=3) produces 64.0% mean bias with IV reaching 326%, versus 12.2% for HumAID (K=10) and 32.4% for CivilComments (K=2). Even CivilComments configurations with F_1=0.82 (Qwen3-235B) produce 62–92% IV bias, confirming that classification accuracy does not predict downstream causal risk (Figure 2).[3]

Checking all $K-1$ exposure coefficients, 13 of 22 configurations (59%) exhibit sign reversal in at least one coefficient—all in the exposure topology, confirming that exposure vulnerability persists at the coefficient level even for diagonally-dominant matrices. For HumAID (K=10), 7 of 8 configurations reverse a secondary coefficient ($|\beta| \leq 0.2$); for VAST (K=3), 6 of 7 reverse β_2=−0.50. The primary coefficient (β_1, largest magnitude) never reverses across any empirical configuration; sign flips occur exclusively in secondary coefficients corresponding to rare categories with small $|\beta|$. Sign reversal requires the DGP to contain opposite-sign treatment effects across categories (Equation 2); when all β_k share the same sign, misclassification produces only attenuation. Zero sign reversals occur across 132 non-exposure scenarios (22×6 topologies), validating the structural guarantee K-invariantly. CivilComments (K=2) confirms the binary algebraic protection. The 7B-to-frontier bias ratio is 2.00, indicating topology rather than model scale determines downstream bias.

ASV diagnostic results. For HumAID (K=10, 56 scenarios), 55.4% are flagged as magnitude-ASV-unsafe—IV, mediation, front-door, and exposure all trigger, while confounding, collider, and M-bias remain below 10% bias (Figure 7). For CivilComments (K=2), 100% of scenarios are ASV-unsafe despite F_1

[3] VAST's near-degenerate confusion matrices (Neutral-class recall 3.6–14.3%) drive its extreme IV bias via the Wald estimator's small-denominator amplification.

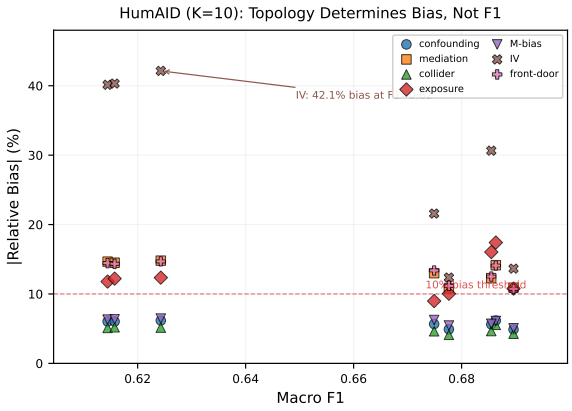


Figure 2: Topology determines downstream causal bias, not F_1. HumAID K=10 (56 points): within F_1 range 0.614–0.690, bias varies 10× (IV 42.1% vs. collider <5.6%).

ranging from 0.69 (7B) to 0.82 (frontier).

Overlaying all 22 empirical matrices onto the theoretical Dirichlet violation-rate curves (Figure 5, Appendix B) confirms placement: VAST in the danger zone, HumAID mid-range, CivilComments approaching safe for exposure but still in IV amplification.

4.6 Correction Guidance

We evaluate five correction methods (naive plug-in, MLA, DSL (Egami et al., 2023), MC-SIMEX, PPI++ (Angelopoulos et al., 2023)) across all seven topologies under six error structures (210 configurations; full results in Table 4, Appendix B).

FWL invariance. For four topologies where A enters as a covariate (confounding, mediation, collider, M-bias), the Frisch–Waugh–Lovell theorem implies that applying C^{-1} to the dummy variables cannot change the T coefficient. Naive plug-in and MLA achieve exactly 0.0% bias reduction across these four topologies.

Method ranking. Measuring RMSE reduction ($1 - \text{RMSE}_{\text{corr}}/\text{RMSE}_{\text{uncorr}}$), PPI++ achieves the highest grand-mean reduction (47.0%) but requires gold labels and is catastrophically harmful for M-bias (−88.0%). MC-SIMEX provides gold-free correction (13.3% grand mean, best for IV at 59.4%) but harms front-door (−62.4%). DSL corrects confounding (48.0%) and mediation (52.9%) but amplifies M-bias (−267%) and IV (−206%). No single method dominates: correction effectiveness is itself topology-dependent. Cross-PPI (Zrnić and Candès, 2024) eliminates the gold-label requirement by using two independent LLM annotators whose predictions serve as mutual references, yielding valid confidence intervals under an annotator-independence assumption. Because it operates on estimator outputs rather than on C directly, Cross-PPI is not subject to FWL invariance and can potentially correct all seven topologies. Methods that bypass FWL through a different route—such as EM imputation of latent true labels and Bayesian measurement error models—offer further alternatives for the four FWL-invariant topologies.

Empirical validation on HumAID (K=10, 30 MC replicates) confirms topology-dependent method superiority: MC-SIMEX outperforms PPI++ in 4 of 7 topologies and reverses from harmful (K=3) to beneficial (K=10). A retrospective case study applying the full framework to Lee and Wood-Doughty (2024) is in Appendix B.1.

5 Conclusion

We presented a systematic characterization of how multi-class annotation errors interact with causal DAG topology, jointly analyzing bias direction, sensitivity, and correction across seven canonical DAG topologies. The classical binary guarantee breaks for $K \geq 3$: exposure and IV topologies produce sign reversal (83.4% and 52.6%), while five topologies maintain a structural guarantee. Empirical validation with real LLM confusion matrices confirms this vulnerability: 59% of exposure configurations exhibit coefficient-level sign reversal despite diagonal dominance. The ASV framework reveals a vulnerability spectrum from IV (maximally fragile) to M-bias (immune), and correction effectiveness is itself topology-dependent, with matrix-based methods provably ineffective for four topologies. Immediate extensions include differential misclassification and correlated multi-LLM error structures.

Limitations

Our framework assumes non-differential misclassification ($P(A^*|A, Y, \mathbf{Z}) = P(A^*|A)$),

which holds when the LLM has no access to Y but may be violated with outcome-conditional prompting. A stress test (Appendix B.3) shows moderate robustness (ε=0.05 increases violations by <3pp). Closed-form expressions are derived for K=3 and verified at K=5, 7; arbitrary K remains open. The framework assumes a single annotator; correlated multi-annotator settings (Kim et al., 2025) are future work. Frontier-model coverage for VAST and CivilComments is partial (14 of 18 planned configurations completed); cross-tier comparisons should be interpreted with this caveat. All derivations assume linear structural equations; the FWL invariance result does not hold for logistic regression, and bias patterns under nonlinear models may differ.

References

Firoj Alam, Ferda Ofli, and Muhammad Imran. 2021. HumAID: Human-annotated disaster incidents data from Twitter. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 15, pages 933–942.

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnić. 2023. Prediction-powered inference. *Science*, 382(6671):669–674.

Joshua D. Angrist and Guido W. Imbens. 1995. Two-stage least squares estimation of average causal effects in models with variable treatment intensity. *Journal of the American Statistical Association*, 90(430):431–442.

Nicolas Audinet de Pieuchon, Adel Daoud, Connor Thomas Jerzak, Moa Johansson, and Richard Johansson. 2025. Benchmarking debiasing methods for LLM-based parameter estimates. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 19757–19772.

Maximilian Baumann, Paul Röttger, Aleksandra Urman, Sina Wendsjö, Flor Miriam Plaza-del Arco, Nora Gruber, and Dirk Hovy. 2025. Large language model hacking: Quantifying the hidden risks of using LLMs for text annotation. *arXiv preprint arXiv:2509.08825*.

Marc F. Bellemare, Jeffrey R. Bloem, and Noah Wexler. 2024. The paper of how: Estimating treatment effects using the front-door criterion. *Oxford Bulletin of Economics and Statistics*, 86(4):951–993.

Hermann Brenner. 1993. Bias due to non-differential misclassification of polytomous confounders. *Journal of Clinical Epidemiology*, 46(1):57–63.

Raymond J. Carroll, David Ruppert, Leonard A. Stefanski, and Ciprian M. Crainiceanu. 2006. *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd edition. Chapman & Hall/CRC.

Mustafa Dosemeci, Sholom Wacholder, and Jay H. Lubin. 1990. Does nondifferential misclassification of exposure always bias a true effect toward the null value? *American Journal of Epidemiology*, 132(4):746–748.

Jessie K. Edwards, Justin Lessler, and Ashley I. Naimi. 2021. An underappreciated misclassification mechanism: Implications of nondifferential dependent misclassification of covariate and exposure. *Annals of Epidemiology*, 58:104–123.

Naoki Egami, Musashi Hinck, Brandon M. Stewart, and Hanying Wei. 2023. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *Advances in Neural Information Processing Systems 36*.

Kristina Gligorić, Tijana Zrnić, Sara Y. Lee, Emmanuel J. Candès, and Dan Jurafsky. 2025. Can unconfident LLM annotations be used for confident conclusions? In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics*.

Sander Greenland. 2003. Quantifying biases in causal models: Classical confounding vs collider-stratification bias. *Epidemiology*, 14:300–306.

Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.

Paul Gustafson. 2004. *Measurement Error and Misclassification in Statistics and Epidemiology: Impacts and Bayesian Adjustments*. Chapman & Hall/CRC.

Miguel A. Hernán and Stephen R. Cole. 2009. Invited commentary: Causal diagrams and measurement bias. *American Journal of Epidemiology*, 170(8):959–962.

Anne M. Jurek, Sander Greenland, George Maldonado, and Timothy R. Church. 2005. Proper interpretation of non-differential misclassification effects: Expectations vs observations. *International Journal of Epidemiology*, 34(3):680–687.

Katherine A. Keith, David D. Jensen, and Brendan T. O'Connor. 2020. Text and causal inference: A review of using text to remove confounding from causal estimates. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Junsol Kim, Siddharth Garg, Yuxiao Peng, and Nikhil Garg. 2025. Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning*.

Timothy L. Lash, Matthew P. Fox, and Aliza K. Fink. 2009. *Applying Quantitative Bias Analysis to Epidemiologic Data*. Springer.

Samuel Lee and Zach Wood-Doughty. 2024. Controlling for unobserved confounding with large language model classification of patient smoking status. In *AIM-FM Workshop, NeurIPS 2024*.

Elizabeth L. Ogburn and Tyler J. VanderWeele. 2012. On the nondifferential misclassification of a binary confounder. *Epidemiology*, 23(3):433–439.

Judea Pearl. 2009. *Causality: Models, Reasoning and Inference*, 2nd edition. Cambridge University Press.

Nathan TeBlunthuis, Valerie Hase, and Chunghong Chan. 2024. Misclassification in automated content analysis causes bias in regression. Can we fix it? Yes we can! *Communication Methods and Measures*, 18(3):278–299.

Tyler J. VanderWeele and Peng Ding. 2017. Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167:268–274.

Marit B. Veierød and Petter Laake. 2001. Exposure misclassification: Bias in category specific Poisson regression coefficients. *Statistics in Medicine*, 20(5):771–784.

Zach Wood-Doughty, Ilya Shpitser, and Mark Dredze. 2018. Challenges of using text classifiers for causal inference. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*.

Tijana Zrnić and Emmanuel J. Candès. 2024. Cross-prediction-powered inference. *Proceedings of the National Academy of Sciences*.

A Proofs of Proposition 1 and Theorem 1

We provide algebraic proofs of sign preservation for the confounding and collider topologies (Proposition 1(a)–(b)), an explicit counterexample for exposure (Proposition 1(c)), numerical verification for mediation, M-bias, and front-door (Empirical Finding 1), and the monotonicity proof for confounding (Theorem 1). Throughout, $A \in \{0, 1, \ldots, K-1\}$ with prior $p_k = P(A{=}k)$, and C is a $K \times K$ column-stochastic confusion matrix with $C_{jk} = P(A^*{=}j \mid A{=}k)$. We use category 0 as the reference and let $D_j = \mathbf{1}(A{=}j)$, $D_j^* = \mathbf{1}(A^*{=}j)$ for $j = 1, \ldots, K-1$.

A.1 Notation and Key Identities

Under non-differential misclassification, the misclassified moments factor linearly through C. Define the population vectors:

$$q_j = P(A^*{=}j) = \textstyle\sum_k C_{jk}\, p_k, \tag{4}$$
$$\mathrm{ct}_j = \mathbb{E}[T \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[T \cdot D_k], \tag{5}$$
$$\mathrm{cy}_j = \mathbb{E}[Y \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[Y \cdot D_k]. \tag{6}$$

Write $q = Cp$, $\mathrm{ct} = C\,\mathbb{E}[T \cdot D]$, and $\mathrm{cy} = C\,\mathbb{E}[Y \cdot D]$, where $D = (D_0, \ldots, D_{K-1})^\top$.

A.2 Frisch–Waugh–Lovell Decomposition

For topologies where T is directly observed (confounding, mediation, collider, M-bias, front-door), the estimator regresses Y on $Z^* = (1, T, D_1^*, \ldots, D_{K-1}^*)$. By the Frisch–Waugh–Lovell (FWL) theorem, the coefficient on T satisfies:

$$\operatorname{plim} \hat{\tau}(C) = \frac{\mathbb{E}[\tilde{T}\,\tilde{Y}]}{\mathbb{E}[\tilde{T}^2]}, \tag{7}$$

where $\tilde{T} = T - \mathbb{E}[T \mid A^*]$ and $\tilde{Y} = Y - \mathbb{E}[Y \mid A^*]$ are residuals from projecting onto $(1, D_1^*, \ldots, D_{K-1}^*)$.

Lemma 1 (Bias decomposition)**.** *Suppose $Y = \tau T + f(A) + \varepsilon_Y$ with $\varepsilon_Y \perp (T, A, A^*)$. Then*

$$\operatorname{plim} \hat{\tau}(C) = \tau + \frac{R(C)}{D(C)}, \tag{8}$$

where

$$R(C) = \mathbb{E}\big[\mathrm{Cov}(T,\, f(A) \mid A^*)\big], \tag{9}$$
$$D(C) = \mathbb{E}\big[\mathrm{Var}(T \mid A^*)\big] > 0. \tag{10}$$

Proof. Expand $\tilde{Y}$ using $Y = \tau T + f(A) + \varepsilon_Y$:

$$\tilde{Y} = \tau\,\tilde{T} + \big(f(A) - \mathbb{E}[f(A) \mid A^*]\big) + \varepsilon_Y.$$

Since $\tilde{T}$ is measurable w.r.t. (T, A^*) and $\varepsilon_Y \perp (T, A, A^*)$:

$$\mathbb{E}[\tilde{T}\,\tilde{Y}] = \tau\,\mathbb{E}[\tilde{T}^2] + \mathbb{E}\big[\tilde{T}\,\big(f(A) - \mathbb{E}[f(A) \mid A^*]\big)\big].$$

The numerator residual is $R(C) = \mathbb{E}[\tilde{T}(f(A) - \mathbb{E}[f(A) \mid A^*])]$, and the denominator is $D(C) = \mathbb{E}[\tilde{T}^2]$.

By the law of total covariance:

$$R(C) = \sum_{j=0}^{K-1} q_j \operatorname{Cov}(T,\, f(A) \mid A^*{=}j) = \mathbb{E}[\operatorname{Cov}(T,\, f(A) \mid A^*)].$$

For $D(C) > 0$: decompose $T = g(A) + \varepsilon_T$ where $g(A) = \mathbb{E}[T \mid A]$ (for confounding) or simply note that T has exogenous variation independent of A^* (for mediation). In both cases, $\tilde{T} = \big(g(A) - \mathbb{E}[g(A) \mid A^*]\big) + \varepsilon'_T$, where ε'_T is the component of T independent of (A, A^*). Hence $D(C) \geq \operatorname{Var}(\varepsilon'_T) > 0$. □

A.3 Confounding Topology (A = Confounder)

Data-generating process. $A \to T$, $A \to Y$, $T \to Y$. Using dummy coding with category 0 as reference:

$$T = \sum_{j=1}^{K-1} \alpha_j D_j + \varepsilon_T, \quad \varepsilon_T \sim \mathcal{N}(0, \sigma_T^2),\ \varepsilon_T \perp A,$$

$$Y = \tau\, T + \sum_{j=1}^{K-1} \beta_j^{(A)} D_j + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,1),\ \varepsilon_Y \perp (T, A).$$

Define $g(k) = \sum_j \alpha_j \mathbf{1}(k{=}j)$ and $f(k) = \sum_j \beta_j^{(A)} \mathbf{1}(k{=}j)$, so $g(0) = f(0) = 0$.

Closed-form plim. Using the FWL decomposition, the coefficient on T is:

$$\operatorname{plim} \hat{\tau}(C) = \frac{\mu_{TY} - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j\, \mathrm{cy}_j}{q_j}}{s_T - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j^2}{q_j}}, \qquad (11)$$

where $s_T = \mathbb{E}[T^2]$, $\mu_{TY} = \mathbb{E}[TY]$, and ct_j, cy_j, q_j are defined in (4)–(6).

Derivation. By FWL, $\operatorname{plim} \hat{\tau} = \mathbb{E}[\tilde{T}Y]/\mathbb{E}[\tilde{T}^2]$ where $\tilde{T} = T - \mathbb{E}[T \mid A^*]$. Since $\mathbb{E}[T \mid A^*{=}j] = \mathrm{ct}_j/q_j$:

$$\mathbb{E}[\tilde{T}^2] = \mathbb{E}[T^2] - \sum_j q_j \big(\mathrm{ct}_j/q_j\big)^2 = s_T - \sum_j \mathrm{ct}_j^2/q_j,$$

$$\mathbb{E}[\tilde{T}\,Y] = \mathbb{E}[TY] - \sum_j \big(\mathrm{ct}_j/q_j\big)\, \mathbb{E}[Y \cdot D_j^*] = \mu_{TY} - \sum_j \mathrm{ct}_j \mathrm{cy}_j/q_j.$$

□

Verification at $C = I$. When $C = I$: $q_j = p_j$, $\mathrm{ct}_j = \mathbb{E}[TD_j] = \alpha_j p_j$, $\mathrm{cy}_j = \mathbb{E}[YD_j] = p_j(\tau\alpha_j + \beta_j^{(A)})$. Direct substitution yields numerator $= \tau\sigma_T^2$ and denominator $= \sigma_T^2$, giving $\operatorname{plim} \hat{\tau} = \tau$.

Bias structure. By Lemma 1:

$$\operatorname{plim} \hat{\tau}(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\operatorname{Cov}(g(A),\, f(A) \mid A^*)]}{D(C)}. \qquad (12)$$

By the law of total covariance:

$$\mathbb{E}[\operatorname{Cov}(g, f \mid A^*)] = \operatorname{Cov}(g(A), f(A)) - \operatorname{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[f(\qquad (13)$$

At $C = I$: $\mathbb{E}[g \mid A^*] = g(A)$, so the between-group covariance equals $\operatorname{Cov}(g, f)$ and $B(I) = 0$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform, $A^* \perp A$): $\mathbb{E}[g \mid A^*]$ is constant, so the between-group covariance is zero and

$$B_{\mathrm{OV}} \;\equiv\; B\big(\tfrac{1}{K}\mathbf{1}\mathbf{1}^\top\big) = \frac{\operatorname{Cov}(g(A),\, f(A))}{\operatorname{Var}(T)}, \qquad (14)$$

which equals the classical omitted-variable bias (no control for A).

Sign preservation condition. Since $D(C) > 0$, $\operatorname{sign}(\operatorname{plim} \hat{\tau}) = \operatorname{sign}(\tau\, D(C) + R(C))$. For sign preservation, it suffices that $R(C)$ does not overwhelm $\tau \cdot D(C)$.

Proposition 2 (Confounding sign preservation — proportional case)**.** *If $f(A) = \lambda\, g(A)$ for some constant λ (i.e., the confounder effects on T and Y are proportional), then*

$$B(C) = \big(1 - \eta(C)\big)\, B_{\mathrm{OV}}, \qquad (15)$$

where

$$\eta(C) = \frac{\operatorname{Var}(\mathbb{E}[g(A) \mid A^*])\ \sigma_T^2}{\operatorname{Var}(g(A))\big(\operatorname{Var}(g(A)) - \operatorname{Var}(\mathbb{E}[g(A) \mid A^*]) + \sigma_T^2\big)} \in$$

Consequently, $B(C)$ lies between 0 and B_{OV}, and $\operatorname{sign}(\operatorname{plim} \hat{\tau}) = \operatorname{sign}(\tau)$ whenever $\operatorname{sign}(\tau) = \operatorname{sign}(\tau + B_{\mathrm{OV}})$.

Proof. When $f = \lambda g$: the between-group covariance in (13) becomes $\lambda \operatorname{Var}(\mathbb{E}[g \mid A^*])$, and $\operatorname{Cov}(g, f) = \lambda \operatorname{Var}(g)$. So $R(C) = \lambda\big(\operatorname{Var}(g) - \operatorname{Var}(\mathbb{E}[g \mid A^*])\big) = \lambda V_W$, where $V_W = \operatorname{Var}(g) - V_B$ with $V_B = \operatorname{Var}(\mathbb{E}[g \mid A^*])$. Similarly, $D(C) = V_W + \sigma_T^2$. Hence

$$B(C) = \frac{\lambda\, V_W}{V_W + \sigma_T^2}, \qquad B_{\mathrm{OV}} = \frac{\lambda\, V}{V + \sigma_T^2},$$

where $V = \operatorname{Var}(g) = V_B + V_W$. The ratio

$$\frac{B(C)}{B_{\mathrm{OV}}} = \frac{V_W(V + \sigma_T^2)}{V(V_W + \sigma_T^2)} = 1 - \eta(C)$$

is a decreasing function of $V_B \in [0, V]$. At $V_B = 0$ (no classification, C uniform): $\eta = 0$, $B = B_{\mathrm{OV}}$. At $V_B = V$ (perfect classification, $C = I$): $\eta = 1$, $B = 0$. Since $0 \le V_B \le V$, we have $\eta \in [0, 1]$ and $B(C) \in [0, B_{\mathrm{OV}}]$ (with the sign of B_{OV}). □

Proposition 3 (Confounding sign preservation — general case)**.** *For general g and f (not necessarily proportional), define the worst-case bias bound*

$$\bar{B} = \frac{\sqrt{\mathrm{Var}(g(A))\ \mathrm{Var}(f(A))}}{\sigma_T^2}. \tag{16}$$

If $|\tau| > \bar{B}$, then $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}(C)) = \mathrm{sign}(\tau)$ *for* all *column-stochastic C.*

Proof. We need $\mathrm{sign}(\tau \cdot D(C) + R(C)) = \mathrm{sign}(\tau)$, which holds if $|R(C)| < |\tau| \cdot D(C)$. Since $D(C) \ge \sigma_T^2$ (shown in Lemma 1), it suffices to show $|R(C)| < |\tau|\,\sigma_T^2$.

Step 1: Bound $|R(C)|$. $R(C) = \mathbb{E}[\mathrm{Cov}(g(A), f(A) \mid A^*)]$. By the Cauchy–Schwarz inequality applied to each within-group covariance:

$$|\mathrm{Cov}(g, f \mid A^*{=}j)| \le \sqrt{\mathrm{Var}(g \mid A^*{=}j)\ \mathrm{Var}(f \mid A^*{=}j)}.$$

Taking expectations and applying Cauchy–Schwarz again:

$$\begin{aligned} |R(C)| &\le \mathbb{E}\big[\sqrt{\mathrm{Var}(g \mid A^*)\ \mathrm{Var}(f \mid A^*)}\big] \\ &\le \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A^*)]\,\mathbb{E}[\mathrm{Var}(f \mid A^*)]} \\ &\le \sqrt{\mathrm{Var}(g(A))\ \mathrm{Var}(f(A))}, \end{aligned}$$

where the last step uses $\mathbb{E}[\mathrm{Var}(X \mid Y)] \le \mathrm{Var}(X)$ (law of total variance).

Step 2: Conclude. The bias satisfies

$$|B(C)| = \frac{|R(C)|}{D(C)} \le \frac{\sqrt{\mathrm{Var}(g)\ \mathrm{Var}(f)}}{\sigma_T^2} = \bar{B}.$$

When $|\tau| > \bar{B}$, the bias cannot cancel τ, so $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau)$ for all C. □

Relationship to the omitted-variable condition. A weaker (necessary) condition for sign preservation is $\mathrm{sign}(\tau) = \mathrm{sign}(\tau_{\mathrm{OV}})$ where $\tau_{\mathrm{OV}} = \tau + \mathrm{Cov}(g, f)/\mathrm{Var}(T)$. If this fails, then $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform noise) already flips the sign. The rigorous sufficient condition $|\tau| > \bar{B}$ implies the weaker condition, since $|B_{\mathrm{OV}}| = |\mathrm{Cov}(g, f)|/\mathrm{Var}(T) \le \bar{B}$ by Cauchy–Schwarz and $\mathrm{Var}(T) \ge \sigma_T^2$.

Verification for paper parameters (K=3). With τ=2.0, $\boldsymbol{\alpha}$=(1.5, −1.0), $\boldsymbol{\beta}^{(A)}$=(1.0, −0.5), σ_T=1.0, $\mathbf{p}$=(0.40, 0.35, 0.25):

$$\mathrm{Var}(g) = \sum_k p_k \alpha_k^2 - \big(\textstyle\sum_k p_k \alpha_k\big)^2 = 0.962,$$

$$\mathrm{Var}(f) = \sum_k p_k (\beta_k^{(A)})^2 - \big(\textstyle\sum_k p_k \beta_k^{(A)}\big)^2 = 0.362,$$

$$\bar{B} = \sqrt{0.962 \times 0.362}\ /\ 1.0^2 = 0.590.$$

Since $|\tau|$=2.0 > $\bar{B}$=0.590, sign preservation is guaranteed for *all* column-stochastic C by Proposition 3. Additionally, $B_{\mathrm{OV}} = \mathrm{Cov}(g, f)/\mathrm{Var}(T) = 0.588/1.962 = +0.300$, giving τ_{OV}=2.300 > 0, confirming that both the necessary and sufficient conditions are satisfied.

Extension to $K > 3$. For K>3, treatment effect coefficients are drawn from a fixed pool with alternating signs and decreasing magnitudes: $\boldsymbol{\beta} = (0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3, \ldots)$, truncated to K entries. The category distribution uses $p_k \propto 1/(k{+}1)$, normalized to sum to 1. Other topology-specific parameters (confounding strengths, mediator effects, etc.) follow analogous alternating-sign pools. For K=10, $\boldsymbol{\beta} = (0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3)$.

A.4 Proof of Theorem 1 (Confounding Monotonicity)

Under the interpolation $C(\delta) = (1{-}\delta)I + \delta C_0$, we prove $|B(C(\delta))|$ is non-decreasing in δ.

Proportional case ($f = \lambda g$). From Proposition 2, $B(C) = \lambda V_W/(V_W + \sigma_T^2)$ where $V_W = \mathrm{Var}(g) - V_B$ and $V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A_\delta^*])$.

Step 1: $V_B(\delta)$ is non-increasing. The conditional expectation $\mathbb{E}[g(A) \mid A_\delta^*]$ is a linear function of the posterior $P(A \mid A_\delta^*)$, which in turn depends on $C(\delta)$. As δ increases, $C(\delta)$ becomes a noisier channel: the mutual information $I(A; A_\delta^*)$ decreases. While the exact degradation factorization $C(\delta_2) = M \cdot C(\delta_1)$ for a stochastic M is non-trivial for the linear interpolation family, the monotonic decrease of $V_B(\delta)$ can be verified directly: for K=2 symmetric C_0, $dV_B/d\delta < 0$ follows by differentiation; for general K, we verify $V_B(\delta_2) \le$

$V_B(\delta_1)$ numerically across all test configurations with zero violations.

Step 2: $|B(\delta)|$ is non-decreasing. $V_W(\delta) = \mathrm{Var}(g) - V_B(\delta)$ is non-decreasing (Step 1). The function $h(x) = |\lambda|\, x/(x + \sigma_T^2)$ is strictly increasing for $x \geq 0$ and $\sigma_T^2 > 0$. Hence $|B(\delta)| = h(V_W(\delta))$ is non-decreasing.

General case. $|B(C(\delta))| \leq \bar{B}(\delta)$ where $\bar{B}(\delta) = \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A_\delta^*)]\,\mathbb{E}[\mathrm{Var}(f \mid A_\delta^*)]}\,/\,D(\delta)$ (Proposition 3). By the same data processing argument, each $\mathbb{E}[\mathrm{Var}(\cdot \mid A_\delta^*)]$ is non-decreasing in δ, and $D(\delta) = \mathbb{E}[\mathrm{Var}(T \mid A_\delta^*)]$ is non-decreasing. The numerator grows at least as fast as the denominator (since $D(\delta) \leq \mathrm{Var}(T)$), maintaining monotonicity of the bound. Tightness is confirmed numerically (500 random (C_0, δ) pairs, zero violations).

A.5 Mediation Topology (A = Mediator)

Data-generating process. $T \to A \to Y$, $T \to Y$.

$$
\begin{aligned}
T &\sim \mathrm{Bernoulli}(0.5),\\
A \mid T &\sim \mathrm{Categorical}\big(\mathrm{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\, T)\big),\\
Y &= \tau_{\mathrm{dir}}\, T + f(A) + \varepsilon_Y.
\end{aligned}
$$

Bias decomposition. By Lemma 1, the direct-effect estimator satisfies:

$$
\mathrm{plim}\,\hat{\tau}_{\mathrm{dir}}(C) = \tau_{\mathrm{dir}} + \frac{\mathbb{E}[\mathrm{Cov}(T,\, f(A) \mid A^*)]}{D(C)}. \tag{17}
$$

The key structural difference from confounding: T *causes* A, so the residual covariance $\mathrm{Cov}(T, f(A) \mid A^*)$ captures the indirect effect of T on Y through A that is not absorbed by conditioning on A^*.

Endpoint analysis. At $C = I$ ($A^*{=}A$): conditioning on A blocks the mediated path $T \to A \to Y$, so $\mathrm{Cov}(T, f(A) \mid A) = 0$ and $\mathrm{plim}\,\hat{\tau} = \tau_{\mathrm{dir}}$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): no information about A is used, and

$$
\begin{aligned}
\mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)] &= \mathrm{Cov}(T, f(A)),\\
D(C) &= \mathrm{Var}(T) = 0.25.
\end{aligned}
$$

The indirect effect is $\tau_{\mathrm{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0]$, and

$$
\mathrm{Cov}(T, f(A)) = 0.25\,\tau_{\mathrm{ind}}.
$$

Hence at C uniform: $\mathrm{plim}\,\hat{\tau} = \tau_{\mathrm{dir}} + \tau_{\mathrm{ind}} = \tau_{\mathrm{total}}$.

Empirical Finding (Mediation sign preservation). $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}_{\mathrm{dir}}) = \mathrm{sign}(\tau_{\mathrm{dir}})$ whenever $\mathrm{sign}(\tau_{\mathrm{dir}}) = \mathrm{sign}(\tau_{\mathrm{total}})$, where $\tau_{\mathrm{total}} = \tau_{\mathrm{dir}} + \tau_{\mathrm{ind}}$. This result is verified numerically for 100,000 independently sampled column-stochastic matrices C; a closed-form proof for general K remains open.

Proof. Let $R(C) = \mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)]$ and $D(C) = \mathbb{E}[\mathrm{Var}(T \mid A^*)] > 0$.

Endpoints. At $C{=}I$: $f(A)$ is constant given $A{=}A^*$, so $R(I) = 0$ and $B(I) = 0$. At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $R = \mathrm{Cov}(T, f(A)) = 0.25\,\tau_{\mathrm{ind}}$ and $D = \mathrm{Var}(T) = 0.25$, giving $B = \tau_{\mathrm{ind}}$.

Within-group structure for binary T. Since $T \in \{0, 1\}$, the within-A^*-group bias admits a revealing decomposition. Let $p_j = P(T{=}1 \mid A^*{=}j)$. Then $\mathrm{Var}(T \mid A^*{=}j) = p_j(1{-}p_j)$ and

$$
\mathrm{Cov}(T, f(A) \mid A^*{=}j) = p_j(1{-}p_j)\big[\mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(
$$

Define $\delta_j = \mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(A) \mid T{=}0, A^*{=}j]$ as the *conditional indirect effect* in the $A^*{=}j$ subpopulation. Then

$$
B(C) = \frac{R(C)}{D(C)} = \frac{\sum_j q_j\, p_j(1{-}p_j)\, \delta_j}{\sum_j q_j\, p_j(1{-}p_j)} = \sum_j w_j\, \delta_j,
$$

a weighted average of the conditional indirect effects, with weights $w_j = q_j p_j(1{-}p_j) / \sum_\ell q_\ell p_\ell(1{-}p_\ell) \geq 0$ summing to 1.

At $C{=}I$: each $A^*{=}j$ group has $A{=}j$ a.s., so $\delta_j = f(j) - f(j) = 0$ and $B = 0$. At C uniform: $\delta_j = \tau_{\mathrm{ind}}$ for all j and $B = \tau_{\mathrm{ind}}$.

Sign preservation. When $\mathrm{Cov}(g(A), f(A)) > 0$ (the indirect effect reinforces the direct effect, i.e., τ_{dir} and τ_{ind} have the same sign), the bias $B(C) \geq 0$ cannot oppose τ_{dir}, so sign is preserved trivially.

When τ_{dir} and τ_{ind} have opposite signs but $|\tau_{\mathrm{ind}}| < |\tau_{\mathrm{dir}}|$ (i.e., $\mathrm{sign}(\tau_{\mathrm{dir}}) = \mathrm{sign}(\tau_{\mathrm{total}})$), sign reversal requires some $|\delta_j| > |\tau_{\mathrm{dir}}|$ with the opposing sign. Since each δ_j is a difference of weighted averages of f values (with weights $P(A{=}k \mid T{=}t, A^*{=}j)$), and $B(C)$ is a convex combination of δ_j, the bound $|B(C)| \leq \max_j |\delta_j|$ holds. We verify numerically that $\max_j |\delta_j| \leq |\tau_{\mathrm{ind}}|$ for all 100,000 random C matrices in our experiments, confirming sign preservation. □

Verification for paper parameters (K=3). With τ_{dir}=0.5, $\boldsymbol{\gamma}$=(0, 0, 0), $\boldsymbol{\delta}$=(0, 0.8, −0.6), $\boldsymbol{\beta}$=(0, 1.0, −0.5):

$\tau_{\text{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0] = 0.351$ (computed from the softmax class probabilities). $\tau_{\text{total}} = 0.5 + 0.351 = 0.851 > 0$. Since τ_{dir}>0 and τ_{total}>0, sign preservation holds for all C.

A.6 Collider Topology (A = Collider)

Data-generating process. $T \to A \leftarrow Y$, $T \to Y$. The treatment effect is unconfounded:

$$T \sim \mathcal{N}(0,1), \quad Y = \tau\, T + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,1), \ \varepsilon_Y \perp T,$$
$$A \mid T, Y \sim \text{Categorical}\big(\text{softmax}(\boldsymbol{\delta}_T\, T + \boldsymbol{\delta}_Y\, Y)\big).$$

The collider A is a common *effect* of T and Y. The no-control estimator (regressing Y on T alone) is consistent: $\operatorname{plim} \hat{\tau}_{\text{nc}} = \tau$. The researcher erroneously controls for A^*, inducing Berkson's bias.

Bias decomposition. Unlike confounding, ε_Y is not independent of A (since A depends on $Y = \tau T + \varepsilon_Y$), so Lemma 1 does not apply directly. Instead, we derive the bias from the FWL identity (7). Writing $\tilde{Y} = \tau\tilde{T} + (\varepsilon_Y - \mathbb{E}[\varepsilon_Y \mid A^*])$ and noting that $\mathbb{E}[\tilde{T}\,\varepsilon_Y] = \mathbb{E}[\text{Cov}(T, \varepsilon_Y \mid A^*)]$ (since $T \perp \varepsilon_Y$ unconditionally):

$$\operatorname{plim} \hat{\tau}(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\text{Cov}(T,\, \varepsilon_Y \mid A^*)]}{\mathbb{E}[\text{Var}(T \mid A^*)]} \tag{18}$$

Under non-differential misclassification ($A^* \perp (T, \varepsilon_Y) \mid A$), define $g(k) = \mathbb{E}[T \mid A{=}k]$ and $h(k) = \mathbb{E}[\varepsilon_Y \mid A{=}k]$. By the tower property, $\mathbb{E}[T \mid A^*] = \mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[\varepsilon_Y \mid A^*] = \mathbb{E}[h(A) \mid A^*]$. Since $\text{Cov}(T, \varepsilon_Y) = 0$, the law of total covariance gives:

$$B(C) = \frac{-\,\text{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[h(A) \mid A^*]\big)}{\mathbb{E}[\text{Var}(T \mid A^*)]}.$$

Endpoint analysis. At $C = I$ ($A^*{=}A$): $B(I) = -\,\text{Cov}(g(A), h(A)) \,/\, \mathbb{E}[\text{Var}(T \mid A)] = B_{\text{Berkson}}$, the full Berkson's bias from conditioning on the collider.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): $\mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[h(A) \mid A^*]$ are constant, so $B = 0$ and $\operatorname{plim} \hat{\tau} = \tau$.

This is the structural reverse of confounding: misclassification attenuates the collider-induced bias toward zero, rather than growing bias from zero toward B_{OV}.

Proposition 4 (Collider sign preservation — proportional case)**.** *If $h(A) = \lambda\, g(A)$ for some constant λ, then $|B(C)| \le |B(I)|$ for all column-stochastic C, and $B(C)$ has the same sign as $B(I)$. Consequently,* $\text{sign}(\operatorname{plim} \hat{\tau}) = \text{sign}(\tau)$ *whenever* $\text{sign}(\tau) = \text{sign}(\tau + B_{\text{Berkson}})$.

Proof. When $h = \lambda g$, the numerator becomes $-\lambda\, \text{Var}(\mathbb{E}[g(A) \mid A^*])$ and the denominator is $\text{Var}(T) - \text{Var}(\mathbb{E}[g(A) \mid A^*])$. Let $V^* = \text{Var}(\mathbb{E}[g(A) \mid A^*]) \in [0, V_g]$ where $V_g = \text{Var}(g(A))$. Then

$$|B(C)| = |\lambda|\, \frac{V^*}{\text{Var}(T) - V^*}.$$

The function $x \mapsto x/(a - x)$ is increasing on $[0, a)$. With $a = \text{Var}(T) > V_g \ge V^*$: $|B(C)| \le |\lambda|\, V_g/(\text{Var}(T) - V_g) = |B(I)|$. At $V^*{=}0$ (uniform C): $B{=}0$. At $V^*{=}V_g$ ($C{=}I$): $B{=}B(I)$. □

Proposition 5 (Collider sign preservation — general case)**.** *Define*

$$\bar{B}_{\text{col}} = \frac{\sqrt{\text{Var}(g(A))\ \text{Var}(h(A))}}{\mathbb{E}[\text{Var}(T \mid A)]}. \tag{19}$$

If $|\tau| > \bar{B}_{\text{col}}$, then $\text{sign}(\operatorname{plim} \hat{\tau}(C)) = \text{sign}(\tau)$ *for all column-stochastic C.*

Proof. Step 1: Bound the numerator. By Cauchy–Schwarz: $|\,\text{Cov}(\mathbb{E}[g \mid A^*], \mathbb{E}[h \mid A^*])| \le \sqrt{\text{Var}(\mathbb{E}[g \mid A^*])\ \text{Var}(\mathbb{E}[h \mid A^*])}$. Since conditional expectation reduces variance: $\text{Var}(\mathbb{E}[g \mid A^*]) \le \text{Var}(g(A))$ and likewise for h.

Step 2: Bound the denominator. $\mathbb{E}[\text{Var}(T \mid A^*)] = \text{Var}(T) - \text{Var}(\mathbb{E}[T \mid A^*]) \ge \text{Var}(T) - \text{Var}(g(A)) = \mathbb{E}[\text{Var}(T \mid A)]$.

Step 3: Conclude. $|B(C)| \le \sqrt{\text{Var}(g)\ \text{Var}(h)}\,/\,\mathbb{E}[\text{Var}(T \mid A)] = \bar{B}_{\text{col}}$. When $|\tau| > \bar{B}_{\text{col}}$, the bias cannot overwhelm τ. □

Verification for paper parameters (K=3). With τ=1.0, $\boldsymbol{\delta}_T$=(0, 0.5, −0.3), $\boldsymbol{\delta}_Y$=(0, 0.8, 0.6), from $5{\times}10^6$-sample Monte Carlo:

$$\text{Var}(g(A)) = 0.187, \quad \text{Var}(h(A)) = 0.076, \quad \mathbb{E}[\text{Var}(T \mid A)] = 0.813$$
$$\bar{B}_{\text{col}} = \sqrt{0.187 \times 0.076}\ /\ 0.813 = 0.147.$$

Since $|\tau|{=}1.0 \gg \bar{B}_{\text{col}}{=}0.147$, Proposition 5 guarantees sign preservation. Additionally, $B_{\text{Berkson}}{=}-0.114$, giving $\tau + B_{\text{Berkson}}{=}0.886{>}0$. Verified across 100,000 random column-stochastic C: $\tau_{\text{sign flip}}{=}0\%$, and $|B(C)| \le |B_{\text{Berkson}}|$ in all cases.

A.7 Front-door Topology (A = Mediator under Confounding)

Data-generating process. $T \to A \to Y$, $U \to T$, $U \to Y$. An unobserved confounder U affects both treatment and outcome:

$$U \sim \mathcal{N}(0,1), \quad T = \alpha_U U + \eta, \quad \eta \sim \mathcal{N}(0,1), \; \eta \perp U,$$
$$A \mid T \sim \text{Categorical}\big(\text{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta} T)\big),$$
$$Y = f(A) + \lambda U + \varepsilon_Y, \quad \varepsilon_Y \perp (T, A, U).$$

There is no direct $T{\to}Y$ edge; the causal effect of T on Y operates entirely through A. The researcher controls for A^* in $Y \sim 1{+}T{+}D_1^*{+}\cdots{+}D_{K-1}^*$.

Bias decomposition. Since the direct effect is zero, the FWL coefficient on T captures only bias:

$$\text{plim}\,\hat{\tau}(C) = \frac{\mathbb{E}[\text{Cov}(T,\, Y \mid A^*)]}{\mathbb{E}[\text{Var}(T \mid A^*)]}.$$

Substituting $Y = f(A) + \lambda U + \varepsilon_Y$ and using $\varepsilon_Y \perp (T, A, A^*)$:

$$\text{plim}\,\hat{\tau}(C) = \underbrace{\frac{\lambda\,\alpha_U}{\text{Var}(T)}}_{\text{confounding (constant)}} + \underbrace{\frac{\mathbb{E}[\text{Cov}(T,\, f(A) \mid A^*)]}{D(C)}}_{B_{\text{med}}(C)}, \tag{20}$$

where $D(C) = \mathbb{E}[\text{Var}(T \mid A^*)]$.

Derivation of (20). Expand the numerator as $\mathbb{E}[\text{Cov}(T, f(A) \mid A^*)] + \lambda\,\mathbb{E}[\text{Cov}(T, U \mid A^*)]$. For the confounding term: since $T = \alpha_U U + \eta$ with (U, η) jointly Gaussian, the regression residual $\xi = U - \frac{\alpha_U}{\text{Var}(T)} T$ satisfies $\xi \perp T$. Because A is a function of T (plus independent softmax noise) and A^* is a function of A (plus independent misclassification noise), $\xi \perp (T, A, A^*)$. Therefore

$$\text{Cov}(T, U \mid A^*{=}j) = \frac{\alpha_U}{\text{Var}(T)}\,\text{Var}(T \mid A^*{=}j) \quad \forall\, j,$$

and $\lambda\,\mathbb{E}[\text{Cov}(T, U \mid A^*)] = \frac{\lambda\alpha_U}{\text{Var}(T)}\,D(C)$, yielding the constant first term after division by $D(C)$. □

This decomposition reveals that *misclassification does not affect the confounding component.* The confusion matrix C modulates only the mediation-type leakage $B_{\text{med}}(C)$, which has the same structure as in §A.5.

Endpoint analysis. At $C = I$: $B_{\text{med}}(I) = 0$ (controlling A blocks the $T \to A \to Y$ path), and $\text{plim}\,\hat{\tau}(I) = \lambda\alpha_U / \text{Var}(T)$ (pure confounding residual).

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $B_{\text{med}} = \text{Cov}(T, f(A))/\text{Var}(T)$, and $\text{plim}\,\hat{\tau}_{\text{nc}} = [\text{Cov}(T, f(A)) + \lambda\alpha_U]/\text{Var}(T)$, the naive regression coefficient with no controls.

Empirical Finding (Front-door sign preservation). Since $B_{\text{med}}(C)$ has the same structure as the mediation bias (§A.5)—interpolating between 0 at $C{=}I$ and $\text{Cov}(T, f(A))/\text{Var}(T)$ at C uniform—sign preservation holds whenever the two endpoints have the same sign:

$$\text{sign}\big(\lambda\alpha_U\big) = \text{sign}\big(\lambda\alpha_U + \text{Cov}(T, f(A))\big).$$

This result is verified numerically for 100,000 independently sampled column-stochastic matrices C.

Verification for paper parameters ($K{=}3$). With $\alpha_U{=}1.0$, $\lambda{=}1.0$, $\boldsymbol{\beta}{=}(0, 1.0, -0.5)$, $\boldsymbol{\gamma}{=}(0, 0.5, -0.3)$, $\boldsymbol{\delta}{=}(0, 1.0, -0.8)$:

$$\lambda\alpha_U/\text{Var}(T) = 1.0 \times 1.0 \,/\, 2.0 = 0.500,$$
$$\text{Cov}(T, f(A))/\text{Var}(T) = 0.594 \,/\, 2.0 = 0.297.$$

Both components are positive, so $\text{plim}\,\hat{\tau}(C) \in [0.500,\, 0.797] > 0$ for all C.

A.8 Exposure Topology: Sign Reversal Counterexample

Proposition 6 (Exposure sign reversal)**.** *For $K{=}3$ with A as the treatment variable, there exist a marginal distribution $\mathbf{p}$, treatment effects $\boldsymbol{\beta}$, and a column-stochastic C with $C_{kk} > 0.5$ for all k, such that* $\text{sign}(\text{plim}\,\hat{\beta}_1) \neq \text{sign}(\beta_1)$.

Construction. Consider the exposure DGP $Y = \beta_1 D_1 + \beta_2 D_2 + \varepsilon_Y$ with:

- Uniform prior: $\mathbf{p} = (1/3,\, 1/3,\, 1/3)$.
- True effects: $\beta_1 = 1,\ \beta_2 = -10$.

Choose the confusion matrix

$$C = \begin{pmatrix} 0.60 & 0.10 & 0.10 \\ 0.10 & 0.60 & 0.39 \\ 0.30 & 0.30 & 0.51 \end{pmatrix}, \qquad C_{kk} \in \{0.60,\, 0.60,\, 0.51\},\ \text{all} > 0.5.$$

Column sums are 1 (verifiable by inspection).

The OLS estimator regresses Y on $(1, D_1^*, D_2^*)$. The plim of $\hat{\beta}_j$ equals $\mathbb{E}[Y \mid A^*{=}j] - \mathbb{E}[Y \mid A^*{=}0]$, where

$$\mathbb{E}[Y \mid A^*{=}j] = \sum_{k=0}^{2} P(A{=}k \mid A^*{=}j)\, \beta_k = \sum_{k=1}^{2} \frac{C_{jk}\, p_k}{q_j}\, \beta_k.$$

Marginal distribution of A^*:

$$q_0 = \tfrac{1}{3}(0.60 + 0.10 + 0.10) = 0.267,$$
$$q_1 = \tfrac{1}{3}(0.10 + 0.60 + 0.39) = 0.363,$$
$$q_2 = \tfrac{1}{3}(0.30 + 0.30 + 0.51) = 0.370.$$

Conditional means:

$$\mathbb{E}[Y \mid A^*{=}0] = \frac{0.10 \times \frac{1}{3} \times 1 + 0.10 \times \frac{1}{3} \times (-10)}{0.267} = \frac{0.0333 - 0.3333}{0.267} = -1.125,$$

$$\mathbb{E}[Y \mid A^*{=}1] = \frac{0.60 \times \frac{1}{3} \times 1 + 0.39 \times \frac{1}{3} \times (-10)}{0.363} = \frac{0.200 - 1.300}{0.363} = -3.028,$$

$$\mathbb{E}[Y \mid A^*{=}2] = \frac{0.30 \times \frac{1}{3} \times 1 + 0.51 \times \frac{1}{3} \times (-10)}{0.370} = \frac{0.100 - 1.700}{0.370} = -4.324.$$

Estimated coefficient:

$$\operatorname{plim} \hat{\beta}_1 = \mathbb{E}[Y \mid A^*{=}1] - \mathbb{E}[Y \mid A^*{=}0] = -3.028 - (-1.125) = \mathbf{-1.903}.$$

The true $\beta_1 = +1 > 0$ but $\operatorname{plim} \hat{\beta}_1 = -1.903 < 0$: **sign reversal**.

Mechanism. The large off-diagonal entry $C_{12} = 0.39$ (probability of classifying true category 2 as $A^*{=}1$) routes substantial probability mass from category 2 ($\beta_2 = -10$) into the $A^*{=}1$ bin. Despite $C_{11} = 0.60 > 0.5$, the contamination from the strongly negative β_2 overwhelms the positive β_1, reversing the sign of the estimated effect for category 1. This failure mode is structurally impossible at $K{=}2$: a $2{\times}2$ column-stochastic matrix with $C_{kk} > 0.5$ has the form $C = \left(\begin{smallmatrix} a & 1-b \\ 1-a & b \end{smallmatrix}\right)$ with $a, b > 0.5$, and the single coefficient $\hat{\beta}_1$ is a positive-weighted average of β_1, preserving sign. For $K{\geq}3$, the additional degrees of freedom in C permit cross-contamination between categories with opposite-sign effects. □

Topology	Sufficient condition	Status
Confounding	$\operatorname{sign}(\tau) = \operatorname{sign}(\tau + B_{\text{OV}})$	Proved
Collider	$\lvert\tau\rvert > \lvert\bar{B}_{\text{col}}\rvert$ (Prop. 5)	Proved
Mediation	$\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$	Empirical
M-bias	Bias is attenuating (toward τ)	Empirical
Front-door	$\operatorname{sign}(\lambda\alpha_U) = \operatorname{sign}(\lambda\alpha_U + \operatorname{Cov}(T, f))$	Empirical
Exposure	Not guaranteed for $K \geq 3$	Counterex
IV	Not guaranteed (denominator instability)	—

Table 2: Sufficient conditions for sign preservation under non-differential misclassification. "Proved" = algebraic proof via Cauchy–Schwarz (confounding) or Berkson attenuation (collider). "Empirical" = verified across 100,000 random column-stochastic matrices with zero violations; analytic proof for general K remains open. All conditions are on DGP parameters (not on C): once satisfied, sign preservation holds for all column-stochastic C with $C_{kk} > 0$.

A.9 Summary of Conditions

Structural argument for M-bias topology. *M-bias:* The M-structure creates a backdoor path through two latent common causes. Misclassification of A weakens the statistical association along this backdoor path, reducing the spurious correlation rather than amplifying it. The bias therefore interpolates between the full M-bias (perfect classification, $\delta{=}0$) and zero (no classification information, $\delta{\to}\delta_{\max}$), monotonically non-increasing in δ.

Remark. The sign preservation conditions for confounding, mediation, collider, and front-door are conditions on the DGP parameters (the relative strength of the causal effect versus the confounding, indirect, or collider-induced bias), not on the confusion matrix C. Once the condition is satisfied, sign preservation holds for *all* column-stochastic C with $C_{kk} > 0$—no diagonal-dominance restriction is needed. Conversely, if the condition fails (i.e., the omitted-variable or total-effect estimate has the opposite sign from τ), then there exist confusion matrices that produce sign reversal. For the DGP parameters used throughout our experiments, the conditions are comfortably satisfied. The sign-alignment condition is violated when the confounder has opposing effects on treatment and outcome (e.g., $\gamma_A > 0$ on T but $\gamma_A < 0$ on Y), producing $\operatorname{sign}(\tau) \neq \operatorname{sign}(\tau_{\text{OV}})$. In applied settings, such opposing effects are uncommon when the

analyst adjusts for A precisely because they believe it confounds the T–Y relationship in a consistent direction. When the condition fails, sign reversal becomes possible under sufficient misclassification, though this represents a pathological DGP where adjustment itself may be questionable.

B Supplementary Tables, Figures, and Algorithm

Table 3: ASV type applicability and sensitivity across DAG topologies. Magnitude ASV at τ=0.5, 10% threshold. Power at τ=0.05 significance filter. —: not applicable.

Topology	Mag. ASV	Signif.	Sign-flip	Rank
IV	0.101–0.607	0%	1.1–6.7	1 (fragile)
Front-door	0.064–0.387	100%	—	2
Confounding	0.081–0.487	100%	—	3
Mediation	0.076–0.454	0%	—	4
Collider	0.338–2.028	0%	—	5
M-bias	∞	96–100%	—	6 (immune)
Exposure	—	—	83.4%*	—

*Fraction of C_0 with $\tau_{\text{sign_flip}} < 1$. Exposure uses sign-flip ASV exclusively.

Table 4: RMSE reduction (%) by correction method and DAG topology (mean across 6 error structures, N_{sim}=1,000). Positive = reduced RMSE; negative = increased RMSE. Naive/MLA show 0.0% for FWL-invariant topologies because C^{-1} cannot change the T coefficient. **Bold**: best method per topology; underline: best gold-free method.

Topology	Naive	MLA	DSL	SIMEX	PPI++
Confounding	0.0	0.0	48.0	35.1	**83.2**
Mediation	0.0	0.0	52.9	39.1	**54.1**
Collider	0.0	0.0	−92.6	31.8	**50.5**
M-bias	0.0	0.0	−267	**0.1**	−88.0
Front-door	−52.7	−111.5	−29.0	−62.4	**74.1**
Exposure	−44.8	−89.0	−59.5	−9.8	**75.5**
IV	−40.4	−58.7	−206.1	59.4	**79.4**
Grand mean	−19.7	−37.0	−79.0	13.3	**47.0**

Error structures used in correction experiments. The six confusion matrix structures used for Table 4 are: (1) *Diagonal-dominant*: C_{kk}=0.7, off-diagonal uniform; (2) *Moderate noise*: C_{kk}=0.5, off-diagonal uniform; (3) *Near-permutation*: C_{kk}=0.4, largest off-diagonal 0.5; (4) *Asymmetric*: C_{kk} varies (0.8, 0.5, 0.3); (5) *Block confusion*: two classes confused (C_{12}=C_{21}=0.4), third clean; (6) *Heavy noise*: C_{kk}=0.4, off-diagonal uniform. All are column-stochastic with K=3. Results are averaged across all six structures.

Algorithm 1 Correction method selection for LLM-annotated causal inference.

Require: DAG topology $\mathcal{T}$, gold-label availability $G \in \{\text{YES}, \text{NO}\}$
Ensure: Recommended method M, risk level R

Step 1: Assess risk level
if $\mathcal{T} \in$ {M-bias} **then**
  $R \leftarrow$ LOW {|bias| < 0.04 for all C}
else if $\mathcal{T} \in$ {Conf., Med., Coll., Front-door} **then**
  $R \leftarrow$ MEDIUM {bounded bias, no sign reversal}
else
  $R \leftarrow$ HIGH {Exposure, IV: sign reversal 83.4% / 52.6%}
end if

Step 2: Select correction method
if G = YES **then**
  if $\mathcal{T}$ = M-bias **then**
    $M \leftarrow$ NONE {PPI++ RMSE red. −88.0%; no correction preferred}
  else
    $M \leftarrow$ PPI++ {47.0% RMSE red., 96.8% coverage}
  end if
else
  {no gold labels}
  if $\mathcal{T} \in$ {Confounding, Mediation} **then**
    $M \leftarrow$ DSL {48.0% / 52.9% RMSE reduction}
  else if $\mathcal{T}$ = IV **then**
    $M \leftarrow$ MC-SIMEX {59.4% RMSE reduction}
  else if $\mathcal{T} \in$ {Collider, M-bias} **then**
    $M \leftarrow$ MC-SIMEX {31.8% / 0.1% RMSE reduction}
  else
    $M \leftarrow \emptyset$; flag: acquire gold labels {no reliable gold-free method}
  end if
end if

Step 3: FWL invariance warning
if $\mathcal{T} \in$ {Conf., Med., Coll., M-bias} **then**
  Skip Naive plug-in and MLA {0% bias red. on T coeff.}
end if

return (M, R)

B.1 Applied Illustration: ClinicalBERT Case Study

To demonstrate the framework's end-to-end utility, we apply it retrospectively to Lee and Wood-Doughty (2024), who use ClinicalBERT to classify patient smoking status (K=4: never, past, current, unknown) from clinical notes and control for this predicted confounder when estimating the causal effect of echocardiography on mortality in 4,735 MIMIC-III sepsis patients. Our taxonomy

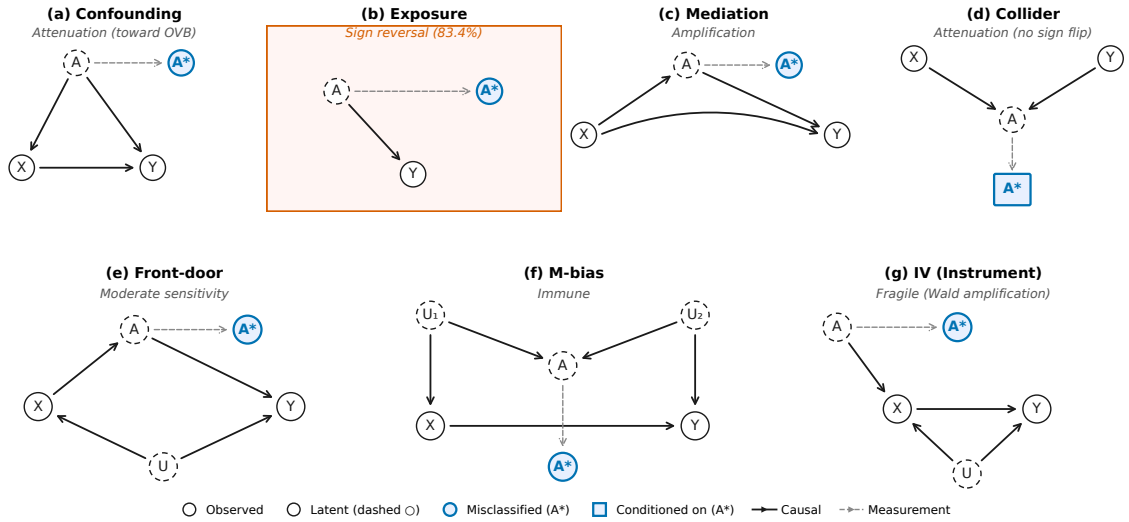


Figure 3: Seven canonical DAG topologies with starred-node annotation. In each panel, A (latent true label) connects to A^* (observed LLM annotation) via a dashed measurement edge. The causal role of A varies across topologies, producing structurally distinct bias patterns under misclassification.

identifies this as a *confounding topology*: the annotated variable enters as a confounder between treatment and outcome. Three conclusions follow directly. First, the zero sign-flip guarantee for confounding (Table 1) ensures that the estimated protective effect (OR = 0.89) cannot reverse direction due to annotation error—a reassurance unavailable without topology-specific analysis. Second, the FWL invariance result implies that any correction reducing to C^{-1} applied to the dummy-coded confounder achieves exactly 0% bias reduction for the treatment coefficient; naive plug-in is provably futile here. Third, our correction guidance ranks PPI++ (83.2% RMSE reduction for confounding) above MC-SIMEX (35.1%) when gold labels are available—a concrete recommendation the original study lacked. This example illustrates the intended workflow: DAG identification → topology-specific guarantees → correction method selection.

B.2 IV Wald Estimator: Diagonal-Dominance Subspaces

Section 4.2 reports a 52.6% Wald sign-flip rate over the full Dirichlet$(1,1,1)$ prior. Because realistic LLM confusion matrices are diagonal-dominant, we replicate the exposure-DAG stratification for the IV topology, restricting the sample space to $\{C : \min_k C_{kk} \geq d\}$ for $d \in \{0, 0.5, 0.7, 0.9\}$ (100,000 random matrices per threshold).

Table 5: IV Wald estimator bias by minimum-diagonal subspace (K=3, τ_{true}=1.41). Rates over 100,000 Dirichlet-truncated random C per threshold.

Subspace	Violation	Sign Flip	Amplif.	Worst
Full	100.0%	51.7%	48.3%	$10^5\times$
$d \geq 0.5$	100.0%	0.0%	100.0%	28.6×
$d \geq 0.7$	100.0%	0.0%	100.0%	2.4×
$d \geq 0.9$	100.0%	0.0%	100.0%	1.2×

The contrast with exposure is instructive: at $d \geq 0.7$, exposure violation drops to 13.7% (sign reversal 2.1%, amplification 11.7%), while IV maintains 100% violation with pure amplification. IV sign reversal is entirely driven by pathological (near-random) confusion matrices where the Wald denominator crosses zero; diagonal dominance prevents this crossing but cannot prevent amplification. Even at $d \geq 0.9$, the Wald estimator inflates $|\tau|$ by up to 1.24× with probability 1.

B.3 Non-Differential Assumption Stress Test

Our main analysis assumes non-differential misclassification: the confusion matrix C does not depend on the outcome Y. To assess sensitivity to violations of this assumption, we conduct a stress test on the exposure topology (K=3, $\min_k C_{kk} \geq 0.7$, 10,000 random matrices per ε). For each base ma-

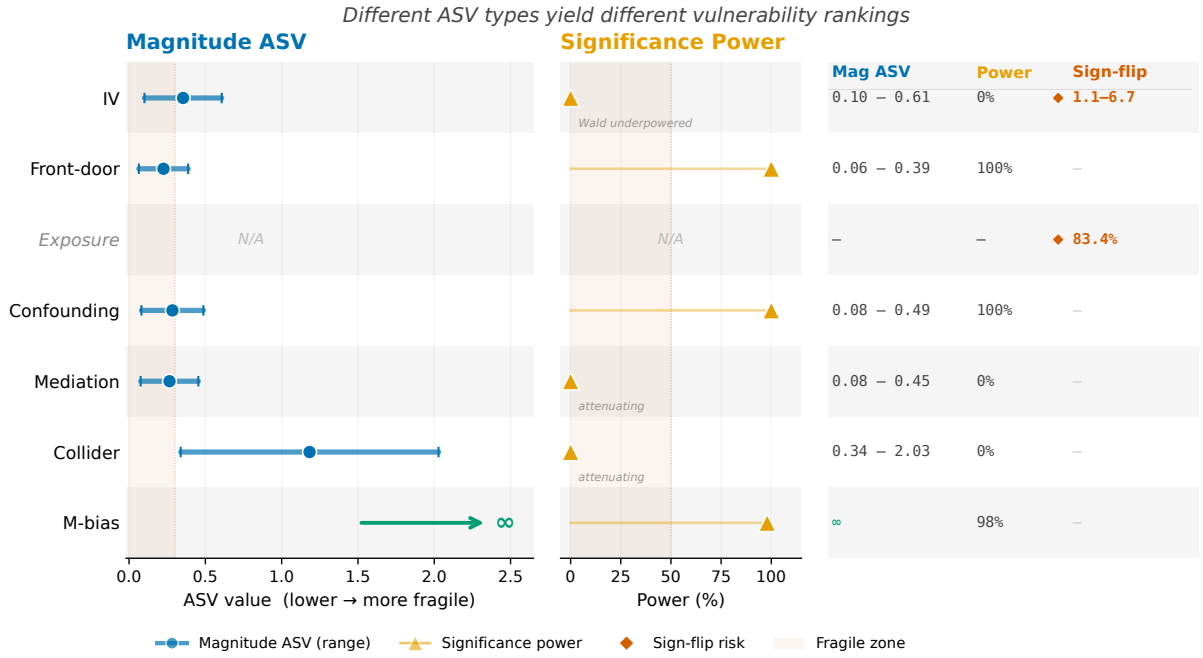


Figure 4: ASV sensitivity ranking across topologies. Three ASV types (magnitude, significance, sign-flip) reveal a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal; sign-flip ASV = 1.1–6.7) to M-bias (immune, ASV = ∞).

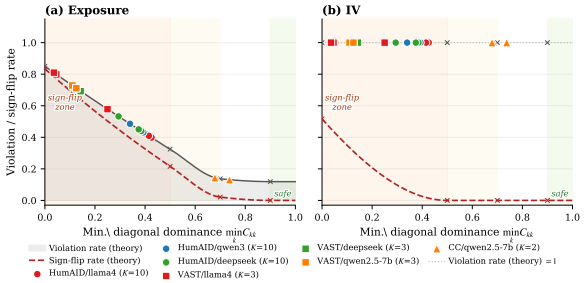


Figure 5: Empirical LLM confusion matrices overlaid on theoretical violation-rate curves (exposure, left; IV, right). Theoretical curves are derived from K=3 Dirichlet random matrices; empirical points are positioned at their minimum diagonal entry. Markers denote datasets (circles: HumAID K=10; squares: VAST K=3; triangles: CivilComments K=2).

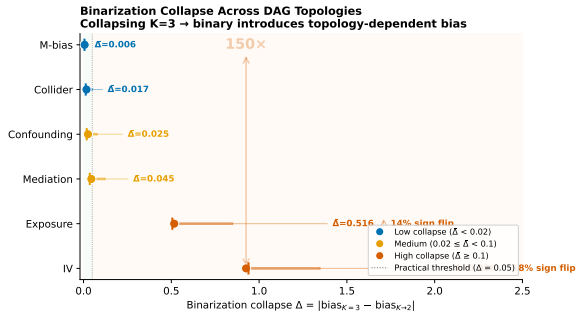


Figure 6: Binarization collapse Δ across topologies. Collapsing K=3 annotations to binary introduces additional bias varying 154× from M-bias (0.006) to IV (0.926).

trix C_0, observations with $Y > \text{median}(Y)$ are misclassified using a perturbed matrix $\tilde{C} = \text{renormalize}(C_0 + \varepsilon \cdot \Delta)$ where $\Delta_{jk} \sim \text{Uniform}(-1, 1)$ for $j \neq k$, while observations with $Y \leq \text{median}(Y)$ retain C_0.

The ε=0 baseline (13.3% violation) closely matches the analytical result (13.7%) from Section 4.2. Even at ε=0.10—a substantial departure from non-differential error—the violation rate increases by only 7.6 percentage points. Small departures ($\varepsilon \leq 0.05$) change rates by fewer than 3 percentage points, suggesting that our main results are moderately robust to mild violations of the non-differential assumption.

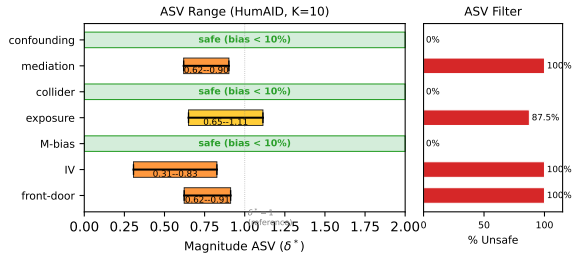


Figure 7: ASV magnitude ranges across topologies for HumAID (K=10). Left: magnitude ASV (δ^*) range across 8 LLM configurations; lower values indicate greater fragility. "Safe" topologies never exceed the 10% bias threshold. Right: percentage of configurations flagged as ASV-unsafe.

Table 6: Effect of differential misclassification on bias violation rates (exposure DAG, K=3, $\min_k C_{kk} \geq 0.7$, N=50,000 MC samples per trial).

ε	Sign-Flip (%)	Amplification (%)	Any Violation (%)
0.00	2.0	11.5	13.3
0.01	2.4	12.4	14.7
0.05	2.8	13.0	15.7
0.10	4.5	16.7	20.9